

Problem Solving in Geometry (IV)

14.10.2023

1. Let O be the circumcentre of an acute triangle ABC . Let M be the midpoint of AC . The straight line BO intersects the altitudes AA_1 and CC_1 at the points H_A and H_C respectively. The circumcircles of the triangles BH_AA and BH_CC have a second point of intersection K . Prove that K lies on the straight line BM . (TOT2021-S-SO3)

2. ABC is an equilateral triangle. Let E , F and K be points such that E lies on side AB , F lies on side AC , K lies on the extension of side AB and $AE = CF = BK$. Let P be the midpoint of segment EF . Prove that the $\angle KPC$ is right angle. (TOT2021-S-JA3)

3. Let M be the midpoint of the side BC of the triangle ABC . The circle ω passes through A , touches the line BC at M , intersects the side AB at the point D and the side AC at the point E . Let X and Y be the midpoints of BE and CD respectively. Prove that the circumcircle of the triangle MXY touches ω . (TOT2021-S-SA3)

4. Two circles α and β with centres A and B respectively intersect at points C and D . The segment AB intersects α and β at points K and L respectively. The ray DK intersects the circle β again at the point N , and the ray DL intersects the circle β for again at the point M . Prove that the intersection point of the diagonals of the quadrangle $KLMN$ coincides with the incentre of the triangle ABC . (TOT2020-F-SA3)

5. Let OP and OQ be the perpendiculars from the circumcentre O of a triangle ABC to the internal and external bisectors of the angle B . Prove that the line PQ divides the segment connecting midpoints of CB and AB into two equal parts. (TOT2019-F-JA4)

6. Let $ABCD$ be a cyclic quadrilateral. Let the circles with diameters AB and CD intersect at two points X_1 and Y_1 , the circles with diameters BC and AD intersect at two points X_2 and Y_2 , the circles with diameters AC and BD intersect at two points X_3 and Y_3 . Prove that the lines X_1Y_1 , X_2Y_2 and X_3Y_3 are concurrent. (TOT2020-S-SA5)